CSC9008 Cyber-Physical Systems
Shake it off: Checkpoint 2

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Outline

Project Overview

Goal

Build a self-stabilizing platform that keeps objects level despite external disturbances (manual tilting, vibration, earthquakes).

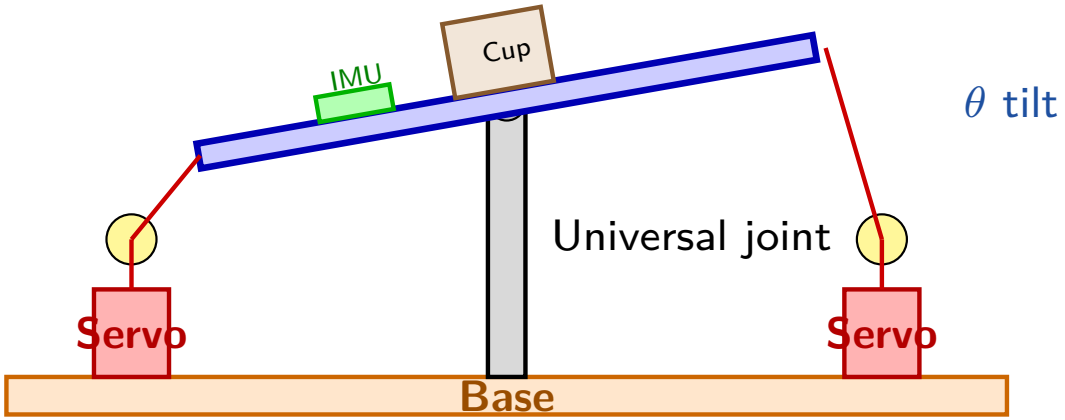
Hardware:

- STM32F401CCU6 (MCU)
- GY-521 / MPU6050 (IMU)
- 4 $\times$ MG90S servos
- Pulleys + strings
- Central universal joint

Control flow:

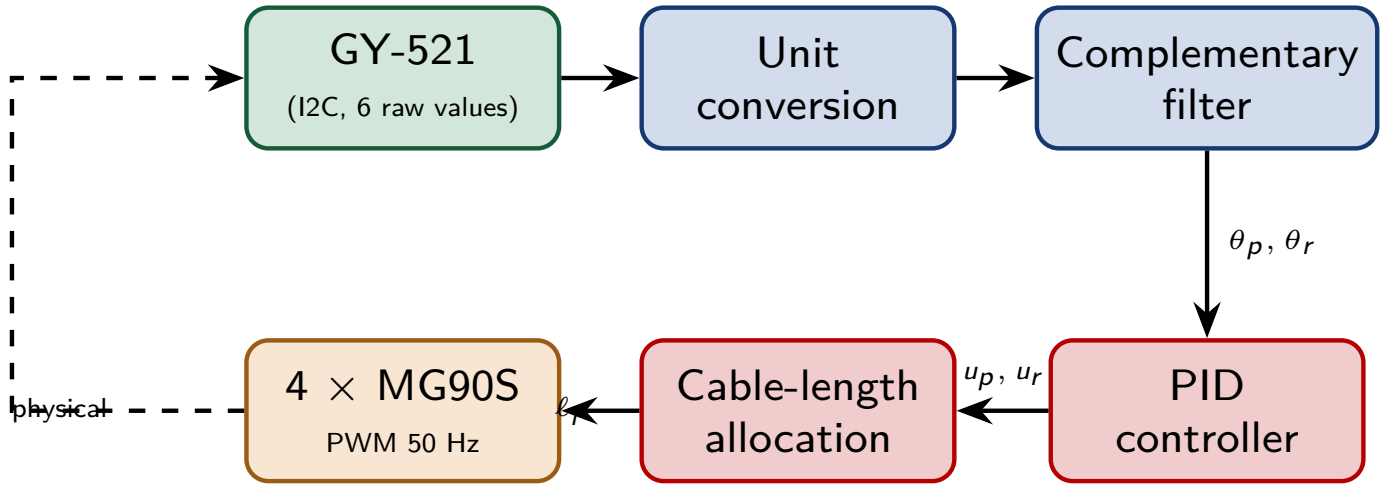
- Sense tilt via IMU
- Compute attitude (pitch, roll)
- PID computes correction
- Servos pull cables
- Platform tilts back to level

Mechanical Structure



- 4 cables pulling 4 platform corners (PA0 back, PA1 left, PA2 right, PA3 front)
- Cross-pairs work antagonistically: one pulls in, the other releases
- Central universal joint bears the load; servos only adjust angle

Signal Flow Diagram



- Closed-loop at **200 Hz** (5 ms control period)
- Pitch and roll axes controlled independently
- All computation runs on STM32F401, USB-CDC for logging

What the MPU6050 Measures

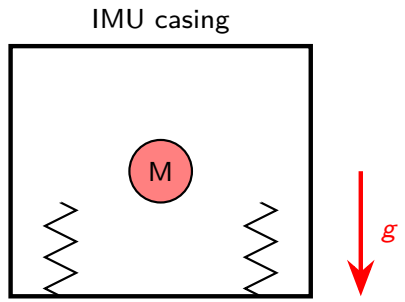
Two sensors in one chip

Accelerometer: measures *specific force* (in g)
Gyroscope: measures *angular rate* (in deg/s)

Accelerometer (counter-intuitive!)
Even when stationary, the chip reads $1\,g$ on z because:

$$\vec{a}_{\text{meas}} = \vec{a}_{\text{motion}} - \vec{g}$$

The spring inside resists gravity $\Rightarrow$ measurement is the reaction force.



I2C Communication and Raw Data Format

Each axis returns a **16-bit signed integer** (range $-32768 \sim +32767$). Six axes packed in 14 bytes starting at register 0x3B:

Reg	0x3B–3C	0x3D–3E	0x3F–40	0x41–42	0x43–44	0x45–46	0x47–48
Data	ACCEL_X	ACCEL_Y	ACCEL_Z	TEMP	GYRO_X	GYRO_Y	GYRO_Z

```
uint8_t buf[14];
HAL_I2C_Mem_Read(&hi2c1, 0xD0, 0x3B, 1, buf, 14, 100);
int16_t ax_raw = (int16_t)((buf[0] << 8) | buf[1]);
int16_t ay_raw = (int16_t)((buf[2] << 8) | buf[3]);
int16_t az_raw = (int16_t)((buf[4] << 8) | buf[5]);
int16_t gx_raw = (int16_t)((buf[8] << 8) | buf[9]);
int16_t gy_raw = (int16_t)((buf[10] << 8) | buf[11]);
int16_t gz_raw = (int16_t)((buf[12] << 8) | buf[13]);
```

Converting Raw Counts to Physical Units

Accelerometer ($\pm 2\,g$ range)

Full scale: $-32768 \leftrightarrow -2g$, $+32767 \leftrightarrow +2g$

$$1 \text{ raw unit} = \frac{2g - (-2g)}{65536} = \frac{1g}{16384}$$

$$\boxed{a_{[g]} = \frac{a_{\text{raw}}}{16384}}$$

Gyroscope (± 250 deg/s range)

$$1 \text{ raw unit} = \frac{500\,\text{deg/s}}{65536} \approx \frac{1\,\text{deg/s}}{131}$$

$$\boxed{\omega_{[\text{deg/s}]} = \frac{\omega_{\text{raw}}}{131}}$$

Sanity check (flat, stationary): $a_x \approx 0$, $a_y \approx 0$, $a_z \approx 1\,g$ (raw ≈ 16384).

Initialization and Calibration

Step 1: Wake up the chip (it boots in sleep mode!)

```
uint8_t d = 0x01;  // X-gyro clock source, wake up
HAL_I2C_Mem_Write(&hi2c1, 0xD0, 0x6B, 1, &d, 1, 100);
// SMPLRT_DIV = 7  -> sample rate 125 Hz
// CONFIG     = 3  -> DLPF cutoff ~44 Hz (Fourier!)
// GYRO_CFG   = 0  -> +/- 250 deg/s
// ACCEL_CFG  = 0  -> +/- 2g
```

Step 2: Static calibration

Average $N = 1000$ samples at rest to estimate the constant bias:

$$a_{\text{off}} = \frac{1}{N}\sum_{k=1}^{N} a_k - \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \qquad \omega_{\text{off}} = \frac{1}{N}\sum_{k=1}^{N} \omega_k$$

Every subsequent reading subtracts this offset:

$$a_{\text{corrected}} = \frac{a_{\text{raw}}}{16384} - a_{\text{off}}$$

Why Fuse Two Sensors?

	Accelerometer	**Gyroscope**
Steady state	Accurate (gravity)	Slow drift
Fast motion	Corrupted by inertia	Accurate
Long term	No drift	Drift grows with t
Good in	Low frequency	High frequency

Key Idea

Use accelerometer for low-frequency reference (gravity is reliable)
Use gyroscope for high-frequency dynamics (no motion artefacts)
Blend them with a **frequency-domain crossover** $\Rightarrow$ Complementary Filter

Kalman filter would adapt the weights dynamically, but the fixed-weight complementary filter is sufficient for our low-speed platform and is far cheaper computationally (1 multiply + 1 add per axis).

Attitude from Accelerometer Alone

Setup: Gravity in the world frame, z-axis up:

$$\vec{g}_W = \begin{bmatrix} 0 \\ 0 \\ -g \end{bmatrix}$$

When the platform rotates by pitch θ_p (around y) and roll θ_r (around x), the IMU sees gravity rotated into its body frame.

Rotation matrix (passive, body-frame view)

$$R(\theta_p, \theta_r) = R_x(\theta_r)\, R_y(\theta_p)$$

$$\vec{g}_{\text{IMU}} = R \cdot \vec{g}_W$$

Since $\vec{g}_W$ has only a z-component, only the third column of R survives.
After multiplication:

$$\begin{aligned} a_x &= \sin\theta_p \\ a_y &= -\sin\theta_r \cos\theta_p \qquad \text{(in units of } g\text{)} \\ a_z &= \cos\theta_r \cos\theta_p \end{aligned}$$

Deriving the Pitch/Roll Formulas

Combine $a_y^2 + a_z^2$ to isolate θ_p:

$$a_y^2 + a_z^2 = \sin^2\theta_r \cos^2\theta_p + \cos^2\theta_r \cos^2\theta_p = \cos^2\theta_p$$

Therefore $\sqrt{a_y^2 + a_z^2} = \cos\theta_p$, and

$$\tan\theta_p = \frac{a_x}{\sqrt{a_y^2 + a_z^2}}$$

Final attitude equations (use atan2 for full-quadrant range)

$$\theta_p^{\text{acc}} = \text{atan2}\left(a_x,\ \sqrt{a_y^2 + a_z^2}\right)$$

$$\theta_r^{\text{acc}} = \text{atan2}(-a_y,\ a_z)$$

Verification (flat): a_x=0, a_y=0, a_z=1 $\Rightarrow$ $\theta_p = \theta_r = 0$. ✓
Verification (pitch 30°): a_x=0.5, a_z=0.866 $\Rightarrow$ $\theta_p = \text{atan2}(0.5, 0.866) = 30°$. ✓

The Gyroscope Path

The gyroscope reports angular rate ω. To get an angle, integrate:

$$\theta^{\text{gyro}}(t) = \theta(0) + \int_0^t \omega(\tau)\, d\tau$$

Discrete form (sample every $\Delta t = 5$ ms):

$$\theta_k^{\text{gyro}} = \theta_{k-1} + \omega_k \cdot \Delta t$$

Drift problem

Any constant bias b in $\omega_{\text{meas}} = \omega_{\text{true}} + b$ grows linearly:

$$\theta_k^{\text{gyro}} = \theta_{\text{true}}(t) + b \cdot t$$

A 0.05 deg/s bias becomes 3° after one minute.

$\Rightarrow$ Gyroscope alone diverges. Accelerometer alone is too noisy.
$\Rightarrow$ We need to fuse them.

The Complementary Filter

Discrete recursion

$$\hat{\theta}_k = \alpha\left(\hat{\theta}_{k-1} + \omega_k \Delta t\right) + (1-\alpha)\,\theta_k^{\text{acc}}$$

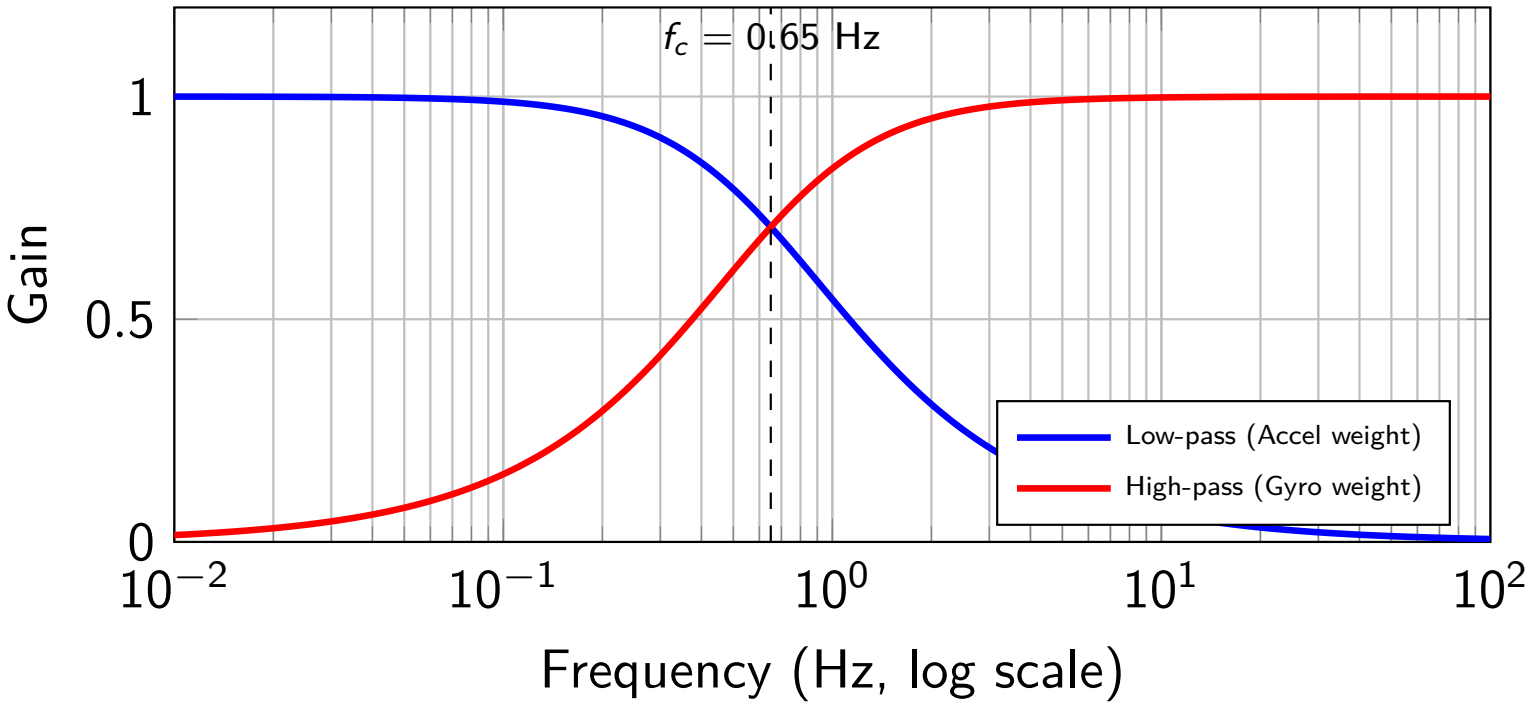
- $\alpha \approx 0.98$: trust gyroscope short-term, use accelerometer to slowly correct drift.
- Equivalent to: **high-pass on gyro + low-pass on accel**, summing to unity gain.

Crossover frequency (Fourier interpretation):

$$f_c = \frac{1-\alpha}{2\pi\,\alpha\,\Delta t} = \frac{0.02}{2\pi \cdot 0.98 \cdot 0.005} \approx 0.65 \text{ Hz}$$

Below 0.65 Hz: accel dominates (gravity reference); above: gyro dominates (fast motion).

Filter Behavior (Conceptual)



- Sum of both curves equals 1 at every frequency $\Rightarrow$ *complementary*
- No phase distortion at the crossover
- One tuning parameter (α), trivial to implement

Implementation

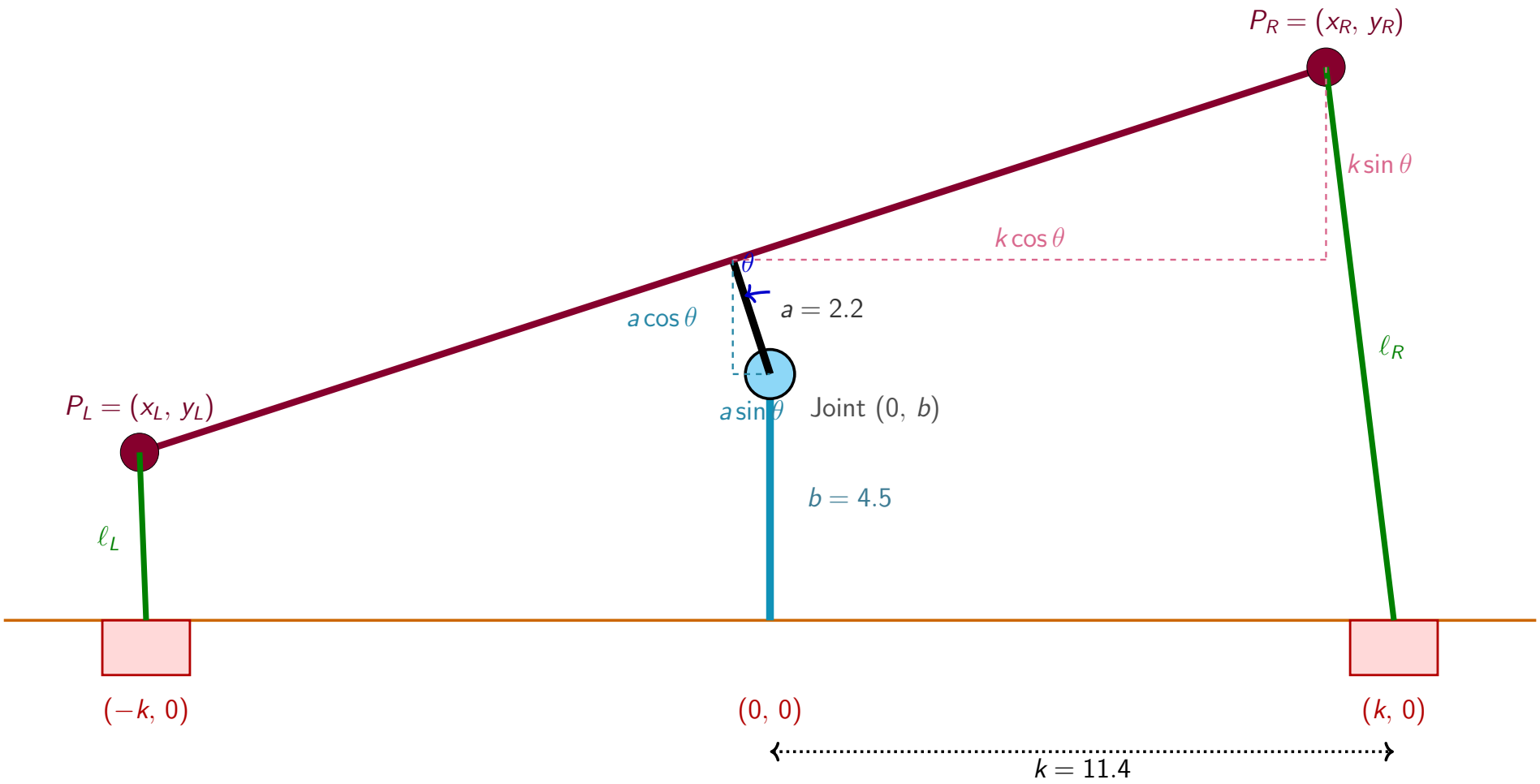
```
#define ALPHA  0.98f
#define DT     0.005f   // 200 Hz control loop
#define R2D    (180.0f / 3.14159265f)

void Attitude_Update(Attitude_t *a, MPU6050_t *imu) {
    // Accel-based static angle
    float pitch_acc = atan2f(-imu->ax,
        sqrtf(imu->ay*imu->ay + imu->az*imu->az)) * R2D;
    float roll_acc  = atan2f(imu->ay, imu->az) * R2D;

    // Complementary filter
    a->pitch = ALPHA * (a->pitch + imu->gx * DT)
             + (1.0f - ALPHA) * pitch_acc;
    a->roll  = ALPHA * (a->roll  + imu->gy * DT)
             + (1.0f - ALPHA) * roll_acc;
}
```

Gyro-to-axis mapping (gx vs gy, and sign) is determined experimentally because it depends on how the IMU is physically oriented on the platform.

Geometric Setup of the Mechanism



Platform rotates about the joint $(0, b)$. The upper pillar (length a) is rigidly attached and tilts with the platform. Cables connect platform ends to fixed servos at $(\pm k, 0)$.

Deriving the Cable Length (Step 1: Locate Platform Center)

The upper pillar rotates rigidly with the platform about the joint at $(0, b)$. The platform *center* is the top of the upper pillar. Before rotation it sits at $(0, b + a)$. The offset from the joint is $(0, a)$. After rotation by θ:

$$\text{offset}' = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ a \end{bmatrix} = \begin{bmatrix} -a\sin\theta \\ a\cos\theta \end{bmatrix}$$

Add back the joint position $(0, b)$:

$$\boxed{P_{\text{center}} = \left(-a\sin\theta,\ b + a\cos\theta\right)}$$

This term is what makes our model different from a naive single-pillar setup. Because a is not negligible compared to k, ignoring it would introduce a position error of up to $a\sin\theta_{\max}$ in the platform center.

Deriving the Cable Length (Step 2: Platform Endpoints)

Each platform endpoint is offset $\pm k$ along the platform direction. After rotation, the offsets become $(\pm k\cos\theta,\ \pm k\sin\theta)$.
Add to the platform center P_{center}:

$$P_R = \big(-a\sin\theta + k\cos\theta,\ b + a\cos\theta + k\sin\theta\big)$$

$$P_L = \big(-a\sin\theta - k\cos\theta,\ b + a\cos\theta - k\sin\theta\big)$$

Cable length is the Euclidean distance from each endpoint to its servo:

$$\ell_R = \|P_R - (k, 0)\|, \quad \ell_L = \|P_L - (-k, 0)\|$$

Result

$$\ell_R(\theta) = \sqrt{(-a\sin\theta + k\cos\theta - k)^2 + (b + a\cos\theta + k\sin\theta)^2}$$

$$\ell_L(\theta) = \sqrt{(-a\sin\theta - k\cos\theta + k)^2 + (b + a\cos\theta - k\sin\theta)^2}$$

Sanity Check at $\theta = 0$

Substituting $\sin 0 = 0$, $\cos 0 = 1$:

$$\ell_R(0) = \sqrt{(0 + k - k)^2 + (b + a + 0)^2} = a + b$$

$$\ell_L(0) = \sqrt{(0 - k + k)^2 + (b + a - 0)^2} = a + b$$

Both cables have length $a + b$ at rest ✓

Cable length *change* relative to neutral position:

$$\Delta\ell_R(\theta) = \ell_R(\theta) - (a + b), \qquad \Delta\ell_L(\theta) = \ell_L(\theta) - (a + b)$$

Numerical example: $a = 2.2$, $b = 4.5$, $k = 11.4$, $\theta = 10^\circ$

$$\ell_R(10^\circ) \approx 8.66 \text{ cm}, \quad \Delta\ell_R \approx +1.96 \text{ cm (release)}$$
$$\ell_L(10^\circ) \approx 4.69 \text{ cm}, \quad \Delta\ell_L \approx -2.01 \text{ cm (reel in)}$$

The two cables move *antagonistically* (one shortens, one lengthens) — exactly the behavior we want.

Control Problem Statement

Objective

Drive the platform attitude (θ_p, θ_r) to $(0, 0)$ in the presence of disturbances.

Define the error signal:

$$e(t) = \theta_{\text{ref}}(t) - \theta(t) = 0 - \theta(t) = -\theta(t)$$

We want a control law $u(t)$ (servo correction in degrees) that:

- Reacts quickly to disturbances
- Eliminates steady-state error (when the table itself is tilted)
- Avoids overshoot and oscillation

$\Rightarrow$ **PID controller**, the workhorse of feedback control.

The Three Terms of PID (Continuous Form)

Continuous form

$$u(t) = \underbrace{K_p\, e(t)}_{\text{Proportional}} + \underbrace{K_i \int_0^t e(\tau)\, d\tau}_{\text{Integral}} + \underbrace{K_d \frac{de(t)}{dt}}_{\text{Derivative}}$$

Term	Reacts to	Cures	Side effect
P	Current error	Slow response	Steady-state error
I	Accumulated error	Steady-state error	Overshoot, slow
D	Rate of change of error	Overshoot, damping	Noise amplification

Analogy:
P = spring (pulls toward target), D = damper (resists motion), I = memory (accumulates over time)

From Discrete Positional to Velocity Form

Approximate the integral by rectangular sum, derivative by backward difference:

$$\int_0^{t_k} e\, d\tau \approx \sum_{i=0}^{k} e_i \Delta t, \qquad \frac{de}{dt} \approx \frac{e_k - e_{k-1}}{\Delta t}$$

Discrete Positional PID

$$u_k = K_p\, e_k + K_i \cdot \Delta t \sum_{i=0}^{k} e_i + K_d \cdot \frac{e_k - e_{k-1}}{\Delta t}$$

Drawback: Requires accumulating historical errors, leading to integral windup.

Discrete Velocity Form PID (Incremental)

Subtracting u_{k-1} from u_k yields the required *change* Δu_k (in our system is the angle change for the next step):

$$\Delta u_k = u_k - u_{k-1} = K_P\,[e_k - e_{k-1}] + K_I\, e_k + K_D\,[e_k - 2e_{k-1} + e_{k-2}]$$

(Note: Constants K_P, K_I, K_D here absorb the Δt and reflect the transformed terms.)

Discrete Velocity Form: PI Implementation

Standard Velocity Form PID: Subtracting positional u_{k-1} from u_k yields the required *change* Δu_k:

$$\Delta u_k = u_k - u_{k-1} = K_p[e_k - e_{k-1}] + K_i e_k + K_d[e_k - 2e_{k-1} + e_{k-2}]$$

Our Velocity Form PI Implementation

To avoid noise amplification from the second difference (D term), our firmware omits the K_d term and implements a **Velocity Form PI** controller:

$$\Delta u_k = K_{P\_vel}\,[e_k - e_{k-1}] + K_{I\_vel}\,e_k$$

Engineering Insight: In microcontrollers, the second difference $[e_k - 2e_{k-1} + e_{k-2}]$ is highly sensitive to sensor noise. Dropping it ensures a smoother control signal.

Velocity Form PI: Mathematical Equivalence

Why does it work? (When Accumulated)
Accumulating (integrating) the velocity step Δu_k over time mathematically reconstructs the standard positional control law:

$$u_k = \sum_{j=0}^{k} \Delta u_j = K_{P\_vel}\, e_k + K_{I\_vel} \sum_{j=0}^{k} e_j$$

- K_{P_vel} **acts as Proportional (P):**
  Multiplies the first difference $[e_k - e_{k-1}]$. When accumulated, the intermediate terms cancel out (telescoping sum), leaving just the current error e_k.
- K_{I_vel} **acts as Integral (I):**
  Multiplies the current error e_k directly. When summed over time, it naturally becomes the accumulated historical error ($\sum e_j$).

Implementation in C: Velocity Form PI Control

In our firmware, we implement the velocity form PI control law directly, where accumulating this step inherently provides a PI response for the platform position.

```
/* 3. Velocity Form PI Calculation */
float err_pitch = 0.0f - att.pitch;

/* KP multiplies the first difference (P); KI multiplies the current error (I) */
float delta_pitch = (KP * (err_pitch - pitch_prev_err)) + (KI * err_pitch);
pitch_prev_err = err_pitch;

/* Expected target mechanical angle */
float next_mech_pitch = target_mech_pitch + delta_pitch;

/* 1. Geometric bounds protection */
if(next_mech_pitch > MAX_MECH_ANGLE) next_mech_pitch = MAX_MECH_ANGLE;
if(next_mech_pitch < MIN_MECH_ANGLE) next_mech_pitch = MIN_MECH_ANGLE;

/* 2. Kinematic Inverse & Anti-Windup Test */
ActuatorTarget pitch_cmd = calculateTargets(next_mech_pitch, a, b, k, C);
float s1_angle = 90.0f + (pitch_cmd.angle_R - base_alpha_R);

if (s1_angle < 0.0f || s1_angle > 180.0f) {
  /* Out of bounds! Discard accumulation (stop integration) to prevent windup */
    pitch_cmd = calculateTargets(target_mech_pitch, a, b, k, C);
} else {
    /* Safe range, officially accumulate */
    target_mech_pitch = next_mech_pitch;
}
```

From PID Output to Servo Action

PID outputs corrections u_p, u_r in **degrees of attitude correction**.
But each servo controls the **length of a cable** that pulls the platform.

Naive approach (does not work well)

Map servo angle directly to PID output: $\theta_{\text{servo}} = 90^\circ + u$.
Problem: 1° of servo rotation does *not* produce 1° of platform tilt. The relationship is highly nonlinear because cables connect a rotating platform to fixed servo positions.

Our approach

Compute the *exact cable length* the platform would need at the desired tilt angle, then convert that length into a servo rotation via the spool radius.

From Cable Length to Servo Angle

Each servo has a spool with circumference C. One full 360° revolution winds/unwinds one full circumference of cable. Therefore:

$$\Delta\ell = C \cdot \frac{\alpha}{360^\circ} \quad \Longrightarrow \quad \boxed{\alpha = \frac{360^\circ \cdot \Delta\ell}{C}}$$

where α is the servo rotation (in degrees) and $\Delta\ell$ is the required cable-length change.

Final servo command (relative to the 90° neutral position):

$$\theta_{\text{servo}} = 90^\circ + \alpha$$

Continuing the example ($\Delta\ell_R = +1.96$ cm, $C = 2\pi r$ with $r = 1.0$ cm $\Rightarrow C \approx 6.28$ cm)

$$\alpha_R = \frac{360 \cdot 1.96}{6.28} \approx 112^\circ$$

A 112° servo rotation is required to release 1.96 cm of cable — which is why bigger spool circumference gives finer (but smaller-range) control.

Full Control Pipeline

1. Read IMU, compute θ_p and θ_r via complementary filter.
2. Run two independent PIDs:

$$u_p = \mathrm{PID}_p(0 - \theta_p), \quad u_r = \mathrm{PID}_r(0 - \theta_r)$$

3. For each axis, compute the required cable lengths using the geometric formulas:

$$\ell_R = \ell_R(u_p), \quad \ell_L = \ell_L(u_p)$$

4. Convert each cable length to a servo rotation:

$$\alpha_i = \frac{360^\circ \cdot \big(\ell_i - (a+b)\big)}{C}$$

5. Output PWM via TIM2 channels (50 Hz, 1.0–2.0 ms pulse width).

Same procedure runs in parallel for the roll axis.
All computations complete within one 5 ms control period.